(e) Determine the volume of CO_2 gas given off when excess $MgCO_3$ is added to 25.0 cm^3 of 0.400 mol / dm^3 HCl at room temperature and pressure.

$$MgCO_3 + 2HCl \rightarrow MgCl_2 + H_2O + CO_2$$

Use the following steps.

- Calculate the number of moles of HCl in 25.0 cm^3 of 0.400 mol / dm^3 of acid.

.... mol

- Determine the number of moles of CO_2 gas given off.

.... mol

- Calculate the volume of CO_2 gas given off in cm^3.

.... cm^3
[3]

[Total: 14]

(c) A student determines the concentration of a solution of dilute sulfuric acid, H_2SO_4, by titration with aqueous sodium hydroxide, NaOH.

step 1 25.0 cm^3 of 0.200 mol / dm^3 NaOH is transferred into a conical flask.

step 2 Three drops of methyl orange indicator are added to the conical flask.

step 3 A burette is filled with H_2SO_4.

step 4 The acid in the burette is added to the conical flask until the indicator changes colour. The volume of acid is recorded. This process is known as titration.

step 5 The titration is repeated several times until a suitable number of results is obtained.

(i) Name the piece of apparatus used to measure exactly 25.0 cm^3 of 0.200 mol / dm^3 NaOH in **step 1**.

.... [1]

(ii) State the colour change of the methyl orange indicator in **step 4**.

from .... to .... [1]

(iii) State how the student decides that a suitable number of results have been obtained.

....

.... [1]

(iv) 20.0 cm^3 of H_2SO_4 reacts with 25.0 cm^3 of 0.200 mol / dm^3 NaOH.

The equation for the reaction is shown.

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

Calculate the concentration of H_2SO_4 using the following steps.

- Calculate the number of moles in 25.0 cm^3 of 0.200 mol / dm^3 NaOH.

.... mol

- Determine the number of moles of H_2SO_4 that react with the NaOH.

.... mol

- Calculate the concentration of H_2SO_4.

.... mol / dm^3
[3]

[Total: 12]

(d) A 25.0 cm^3 sample of limewater is placed in a conical flask. The concentration of $Ca(OH)_2$ in the limewater is determined by titration with dilute hydrochloric acid, HCl.

(i) Name the item of apparatus used to measure the volume of acid in this titration.

.... [1]

(ii) State the type of reaction which takes place.

.... [1]

(iii) As well as limewater and dilute hydrochloric acid, state what other type of substance must be added to the conical flask.

.... [1]

(iv) The equation for the reaction is shown.

$$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$$

20.0 cm^3 of 0.0500 mol / dm^3 HCl reacts with the 25.0 cm^3 of $Ca(OH)_2$.

Determine the concentration of $Ca(OH)_2$ in g / dm^3. Use the following steps.

- Calculate the number of moles in 20.0 cm^3 of 0.0500 mol / dm^3 HCl.

.... mol

- Determine the number of moles of $Ca(OH)_2$ in 25.0 cm^3 of the limewater.

.... mol

- Calculate the concentration of $Ca(OH)_2$ in mol / dm^3.

.... mol / dm^3

- Determine the concentration of $Ca(OH)_2$ in g / dm^3.

.... g / dm^3
[5]

[Total: 21]

(c) The formula for crystals of hydrated copper(II) sulfate is $CuSO_4{\bullet}5H_2O$.

Hydrated copper(II) sulfate is made by reacting copper(II) oxide with dilute sulfuric acid.

The overall equation is shown.

$$CuO + H_2SO_4 + 4H_2O \rightarrow CuSO_4{\bullet}5H_2O$$

The crystals are made using the following steps:

step 1 50.0 cm^3 of 0.200 mol / dm^3 dilute sulfuric acid is heated in a beaker. Powdered copper(II) oxide is added until the copper(II) oxide is in excess. Aqueous copper(II) sulfate is formed.

step 2 The excess copper(II) oxide is separated from the aqueous copper(II) sulfate.

step 3 The aqueous copper(II) sulfate is heated until a saturated solution is formed.

step 4 The saturated solution is allowed to cool and crystallise.

step 5 The crystals are removed and dried.

Calculate the maximum mass of copper(II) sulfate crystals, $CuSO_4{\bullet}5H_2O$, that can form using the following steps.

- Calculate the number of moles of H_2SO_4 in 50.0 cm^3 of 0.200 mol / dm^3 H_2SO_4.

.... mol

- Deduce the number of moles of $CuSO_4{\bullet}5H_2O$ that can form.

.... mol

- The M_r of $CuSO_4{\bullet}5H_2O$ is 250.

  Calculate the maximum mass of $CuSO_4{\bullet}5H_2O$ that can form.

.... g

[3]

(h) Some salts can be made by titration.

In a titration experiment, 20.0 cm^3 of aqueous sodium hydroxide reacts exactly with 25.0 cm^3 of 0.100 mol / dm^3 dilute sulfuric acid to make sodium sulfate.

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

(i) Circle the name of the type of reaction that takes place.

decomposition **neutralisation** **precipitation** **reduction**

[1]

(ii) Calculate the concentration of the aqueous sodium hydroxide in g / dm^3 using the following steps.

- Calculate the number of moles of dilute sulfuric acid used.

.... mol

- Determine the number of moles of sodium hydroxide which react with the dilute sulfuric acid.

.... mol

- Calculate the concentration of the aqueous sodium hydroxide in mol / dm^3.

.... mol / dm^3

- Calculate the concentration of the aqueous sodium hydroxide in g / dm^3.

.... g / dm^3

[5]

[Total: 17]

(ii) In an experiment, 1.61 g of $Na_2SO_4 \cdot \boldsymbol{x}H_2O$ is heated until all the water is given off. The mass of Na_2SO_4 remaining is 0.71 g.

[M_r: Na_2SO_4,142; H_2O,18]

Determine the value of ***x*** using the following steps.

- Calculate the number of moles of Na_2SO_4 remaining.

.... mol

- Calculate the mass of H_2O given off.

.... g

- Calculate the number of moles of H_2O given off.

.... mol

- Determine the value of ***x***.

x = ...
[4]

[Total: 15]

7 Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **V** has the following composition by mass.

C, 48.65%; H, 8.11%; O, 43.24%

Calculate the empirical formula of compound **V**.

empirical formula = ... [3]

(b) Compound **W** has the empirical formula CH_4O and a relative molecular mass of 32.

Calculate the molecular formula of compound **W**.

molecular formula = ... [1]

(c) Compounds **X** and **Y** have the same general formula.

X and **Y** are both carboxylic acids.

Compound **X** has the molecular formula $C_2H_4O_2$.

Compound **Y** has the molecular formula $C_4H_8O_2$.

(i) Deduce the general formula of compounds **X** and **Y**.

.... [1]

2 Silver has an atomic number of 47.

(a) Naturally occurring atoms of silver are ^{107}Ag and ^{109}Ag.

(i) State the name given to atoms of the same element with different nucleon numbers.

.... [1]

(ii) Complete the table to show the number of protons, neutrons and electrons in each atom and ion of silver shown.

	$^{107}_{47}Ag$	$^{109}_{47}Ag^{+}$
protons		
neutrons		
electrons		

[3]

(iii) Complete this definition of relative atomic mass.

Relative atomic mass is the .... mass of naturally occurring atoms of an element on a scale where the .... atom has a mass of exactly .... units.

[3]

(iv) A sample of silver has a relative atomic mass of 108.0.

Deduce the percentage of ^{107}Ag present in this sample of silver.

.... [1]

(b) Silver nitrate is a salt of silver made by reacting silver oxide with an acid.

Write the formula of the acid which reacts with silver oxide to form silver nitrate.

.... [1]

3 Sodium hydrogencarbonate is found in baking powder.

When sodium hydrogencarbonate is heated it forms three products.

$$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

(a) Name the type of reaction that takes place when sodium hydrogencarbonate reacts in this way.

.... [1]

(b) Calculate the volume of carbon dioxide formed at room temperature and pressure when 12.6 g of $NaHCO_3$ is heated using the following steps:

- determine the mass of one mole of $NaHCO_3$

.... g

- calculate the number of moles of $NaHCO_3$ used

.... moles

- determine the number of moles of carbon dioxide formed

.... moles

- calculate the volume of carbon dioxide formed at room temperature and pressure.

.... dm^3
[4]

(c) Limewater is aqueous calcium hydroxide. Carbon dioxide turns limewater milky because a white precipitate forms.

Write the formula of:

- calcium hydroxide ....
- the white precipitate that forms when limewater turns milky. ....

[2]

[Total: 7]

(c) Some chlorides are hydrated.

When hydrated barium chloride crystals, $BaCl_2 \cdot \boldsymbol{x}H_2O$, are heated they give off water.

$$BaCl_2 \cdot \boldsymbol{x}H_2O(s) \rightarrow BaCl_2(s) + \boldsymbol{x}H_2O(g)$$

A student carries out an experiment to determine the value of ***x*** in $BaCl_2 \cdot \boldsymbol{x}H_2O$.

step 1 Hydrated barium chloride crystals are weighed.

step 2 The hydrated barium chloride crystals are then heated.

step 3 The remaining solid is weighed.

(i) Describe how the student can be sure that all the water is given off.

....

....

.... [2]

(ii) In an experiment, 4.88 g of $BaCl_2 \cdot \boldsymbol{x}H_2O$ is heated until all the water is given off. The mass of $BaCl_2$ remaining is 4.16 g.

[M_r: $BaCl_2$, 208; H_2O, 18]

Determine the value of ***x*** using the following steps.

- Calculate the number of moles of $BaCl_2$ remaining.

.... mol

- Calculate the mass of H_2O given off.

.... g

- Calculate the number of moles of H_2O given off.

.... mol

- Determine the value of ***x***.

x = ...
[4]

[Total: 15]

7 Many organic compounds contain carbon, hydrogen and oxygen only.

(a) An organic compound **R** has the following composition by mass.

C, 69.77%; H, 11.63%; O, 18.60%

Calculate the empirical formula of compound **R**.

empirical formula = ... [2]

(b) Compound **S** has the empirical formula CH_2O and a relative molecular mass of 60.

Calculate the molecular formula of compound **S**.

molecular formula = ... [2]

(c) Compounds **T** and **V** have the same molecular formula, $C_3H_6O_2$.

- Compound **T** is an ester.
- Compound **V** contains a –COOH functional group.

(i) State the name given to compounds with the same molecular formula but different structures.

.... [1]

(ii) Name the homologous series that **V** is a member of.

.... [1]

2 This question is about copper and its compounds.

(a) Copper has two different naturally occurring atoms, ^{63}Cu and ^{65}Cu.

(i) State the term used for atoms of the same element with different nucleon numbers.

.... [1]

(ii) The atomic number of copper is 29.

Complete the table to show the number of protons, neutrons and electrons in the particles of copper shown.

	^{63}Cu	$^{65}Cu^{2+}$
protons		
neutrons		
electrons		

[3]

(iii) Relative atomic mass is the average mass of naturally occurring atoms of an element.

The percentage of the naturally occurring atoms in a sample of copper is shown.

^{63}Cu	^{65}Cu
70%	30%

Deduce the relative atomic mass of copper in this sample.

Give your answer to **one** decimal place.

relative atomic mass = ... [2]

(d) When solid copper(II) nitrate is heated copper(II) oxide, nitrogen dioxide and oxygen are formed.

$$2Cu(NO_3)_2 \rightarrow 2CuO + 4NO_2 + O_2$$

Calculate the volume of nitrogen dioxide formed at room temperature and pressure when 4.7 g of $Cu(NO_3)_2$ is heated.

Use the following steps:

- calculate the mass of one mole of $Cu(NO_3)_2$

.... g

- calculate the number of moles of $Cu(NO_3)_2$ used

.... moles

- determine the number of moles of nitrogen dioxide formed

.... moles

- calculate the volume of nitrogen dioxide formed at room temperature and pressure.

.... dm^3
[4]

(e) Write the chemical equation to show the action of heat on sodium nitrate, $NaNO_3$.

.... [2]

[Total: 22]

2 Acids are important laboratory chemicals.

(a) Some acids completely dissociate in water to form ions.

(i) State the term applied to acids that completely dissociate in water.

.... [1]

(ii) Complete the equation to show the complete dissociation of sulfuric acid in water.

$H_2SO_4 \rightarrow$ .... [2]

(iii) State the colour of methyl orange in sulfuric acid.

.... [1]

(b) The equation for the reaction between powdered zinc carbonate and dilute nitric acid is shown.

$ZnCO_3$....... + $2HNO_3$....... $\rightarrow$ $Zn(NO_3)_2$....... + H_2O...... + CO_2.......

(i) Complete the equation by adding state symbols. [2]

(ii) A student found that 2.5 g of zinc carbonate required 20 cm^3 of dilute nitric acid to react completely.

Calculate the concentration of dilute nitric acid using the following steps:

- calculate the mass of 1 mole of $ZnCO_3$

.... g

- calculate the number of moles of $ZnCO_3$ reacting

.... moles

- determine the number of moles of HNO_3 reacting

.... moles

- calculate the concentration of HNO_3.

.... mol / dm^3
[4]

[Total: 10]

3 Lead is a metallic element in Group IV. One of the ores of lead is galena, which is an impure form of lead(II) sulfide, PbS.

Lead also occurs in the ore cerussite, which contains lead(II) carbonate, $PbCO_3$.

(a) Calculate the relative formula mass, M_r, of $PbCO_3$.

M_r of $PbCO_3$ = ... [1]

(b) The M_r of PbS is 239.

Calculate the percentage of lead by mass in PbS.

percentage of lead by mass in PbS = ... [1]

(c) The percentage of lead by mass in $PbCO_3$ is 77.5%.

Use this information and your answer to **(b)** to suggest whether it would be better to extract lead from $PbCO_3$ or PbS.

Give a reason for your answer.

....

.... [1]

(d) When lead(II) carbonate is heated it decomposes into lead(II) oxide, PbO, and carbon dioxide.

Write a chemical equation for this reaction.

.... [1]

(e) Lead(II) carbonate reacts with dilute nitric acid. One of the products is aqueous lead(II) nitrate, $Pb(NO_3)_2$.

Write a chemical equation for this reaction.

.... [2]

4 Carbon is an important element.

(a) Carbon exists as the isotopes $^{12}_{6}C$ and $^{13}_{6}C$.

Complete the table.

isotope	number of protons in one atom	number of electrons in one atom	number of neutrons in one atom
$^{12}_{6}C$			
$^{13}_{6}C$			

[2]

(b) Name **two** forms of the element carbon which have giant covalent structures.

.... and .... [1]

(c) The Avogadro constant is the number of particles in 1 mole.

The numerical value of the Avogadro constant is 6.02×10^{23}.

(i) Calculate the number of molecules in 22.0 g of carbon dioxide, CO_2.

.... molecules [2]

(ii) Calculate the number of molecules in 6.00 dm^3 of carbon dioxide gas at room temperature and pressure.

.... molecules [1]

[Total: 6]

(e) PCl_5 reacts with lithium fluoride, LiF, to form $LiPF_6$.

$$PCl_5 + 6LiF \rightarrow LiPF_6 + 5LiCl$$

Calculate the mass of LiF needed to form 3.04 g of $LiPF_6$ using the following steps.

- Calculate the number of moles of $LiPF_6$ formed.
  [M_r: $LiPF_6$, 152]

number of moles = ...

- Deduce the number of moles of LiF needed.

number of moles = ...

- Calculate the mass of LiF needed.

mass = ... g
[3]

(f) Lithium fluoride has ionic bonding.

(i) What is an ionic bond?

....

.... [2]

(ii) Give **two** physical properties of ionic compounds.

....

.... [2]

[Total: 28]

1 This question is about elements **X**, **Y** and **Z**.

(a) An atom of element **X** is represented as ${}^{34}_{16}\mathbf{X}$.

(i) Name the different types of particles found in the nucleus of this atom of **X**.

....

.... [2]

(ii) What is the term for the total number of particles in the nucleus of an atom?

.... [1]

(iii) What is the total number of particles in the nucleus of an atom of ${}^{34}_{16}\mathbf{X}$?

.... [1]

(iv) What is the electronic structure of the ion $\mathbf{X}^{2-}$?

.... [1]

(v) Suggest the formula of the compound formed between aluminium and **X**.

.... [1]

(b) **(i)** What term is used to describe atoms of the same element with different numbers of particles in the nucleus?

.... [1]

(ii) Identify the atom against which the relative masses of all other atoms are compared.

.... [1]

(iii) What is the name of the amount of any substance that contains 6.02×10^{23} particles?

.... [1]

(iv) The constant 6.02×10^{23} has a name.

What is the name of this constant?

.... [1]

(c) Part of the definition of relative atomic mass is 'the average mass of naturally occurring atoms of an element'.

Some relative atomic masses are not whole numbers.

Element **Y** has only two different types of atom, $^{69}\mathbf{Y}$ and $^{71}\mathbf{Y}$.

The ratio of atoms present in element **Y** is shown.

$$^{69}\mathbf{Y} : {}^{71}\mathbf{Y} = 3 : 2$$

- Calculate the relative atomic mass of element **Y** to **one decimal place**.

relative atomic mass = ...

- Identify element **Y**.

.... [3]

(d) Element **Z** is in Period 3 and Group V.

(i) Identify element **Z**.

.... [1]

(ii) Explain in terms of electron transfer why **Z** behaves chemically as a non-metal.

....

.... [2]

[Total: 16]

(c) Aqueous sodium hydroxide, NaOH(aq), is a strong alkali that reacts with dilute sulfuric acid exothermically.

(i) What type of reaction is this?

.... [1]

(ii) Complete the equation for the reaction between aqueous sodium hydroxide and dilute sulfuric acid.

$2NaOH + H_2SO_4 \rightarrow$ .... + ...

[2]

(d) A student wanted to find the concentration of some dilute sulfuric acid by titration. The student found that 25.0 cm^3 of 0.0400 mol / dm^3 NaOH(aq) reacted exactly with 20.0 cm^3 of H_2SO_4(aq).

(i) Name a suitable indicator to use in this titration.

.... [1]

(ii) Calculate the concentration of the H_2SO_4(aq) in mol / dm^3 using the following steps.

- Calculate the number of moles of NaOH in 25.0 cm^3.

moles = ...

- Deduce the number of moles of H_2SO_4 that reacted with the 25.0 cm^3 of NaOH(aq).

moles = ...

- Calculate the concentration of H_2SO_4(aq) in mol / dm^3.

concentration = ... mol / dm^3
[3]

(iii) Calculate the concentration of the 0.0400 mol / dm^3 NaOH(aq) in g / dm^3.

concentration = ... g / dm^3 [2]

[Total: 16]

(d) In terms of particles, explain what happens to the rate of this reaction when the temperature is increased.

....

....

....

....

.... [3]

(e) The equation for the decomposition of hydrogen peroxide is shown.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

$25.0\,cm^3$ of aqueous hydrogen peroxide forms $48.0\,cm^3$ of oxygen at room temperature and pressure (r.t.p.).

Calculate the concentration of aqueous hydrogen peroxide at the start of the experiment using the following steps.

- Calculate the number of moles of oxygen formed.

.... mol

- Deduce the number of moles of hydrogen peroxide that decomposed.

.... mol

- Calculate the concentration of hydrogen peroxide in mol/dm^3.

.... mol/dm^3
[3]

(f) Oxygen can also be produced by the decomposition of potassium chlorate(V), $KClO_3$.

The only products of this decomposition are potassium chloride and oxygen.

Write a chemical equation for this decomposition.

.... [2]

[Total: 16]

(iv) The equation for the reaction in **stage 3** is shown.

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

Calculate the volume of O_2 gas, at room temperature and pressure (r.t.p.), needed to produce 1260 g of HNO_3.
Use the following steps.

- Calculate the number of moles of HNO_3.

moles of HNO_3 = ...

- Deduce the number of moles of O_2 that reacted.

moles of O_2 = ...

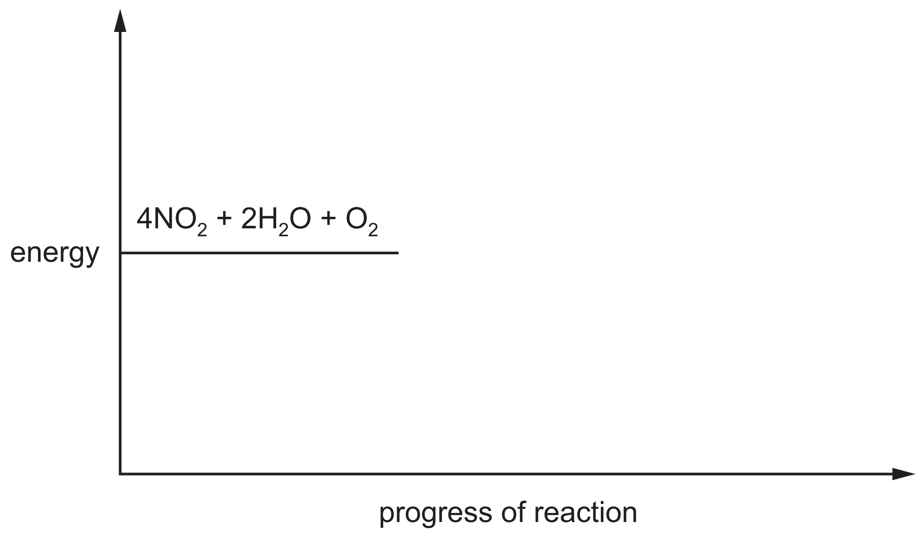
- Calculate the volume of O_2 gas that reacts at room temperature and pressure (r.t.p.).

volume of O_2 gas = ... dm^3
[4]

(e) The reaction in **stage 3** is exothermic.

$$4NO_2 + 2H_2O + O_2 \rightarrow 4HNO_3$$

Complete the energy level diagram for this reaction. Include an arrow that clearly shows the energy change during the reaction.



[3]

[Total: 18]

(f) A compound contains 85.7% carbon and 14.3% hydrogen by mass.

(i) Calculate the empirical formula of this compound.

Show your working.

.... [2]

(ii) The molecular mass of the compound is 112.

Calculate the molecular formula of this compound.

.... [1]

[Total: 16]

(c) On analysing the crystals, the student found that one mole of the hydrated iron(II) sulfate crystals, $FeSO_4 \cdot xH2O$, had a mass of 278 g.

Determine the value of x using the following steps:

- calculate the mass of one mole of $FeSO_4$

mass = ... g

- calculate the mass of H_2O present in one mole of $FeSO_4 \cdot xH_2O$

mass of H_2O = ... g

- determine the value of x.

x = ...
[3]

(d) Insoluble salts can be made by mixing solutions of two soluble salts.

A student followed the procedure shown to make silver bromide, an insoluble salt.

step 1 Add aqueous silver nitrate to a beaker. Then add aqueous potassium bromide and stir.

step 2 Filter the mixture formed in **step 1**.

step 3 Dry the residue.

(i) State the term used to describe this method of making salts.

.... [1]

(ii) Give the observation the student would make during **step 1**.

.... [1]

(iii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous potassium bromide.

Include state symbols.

.... [3]

(e) Sodium chloride is an ionic salt. It can be made by reacting sodium with chlorine gas.

The equation for this reaction is shown.

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

Calculate the volume of chlorine gas, in cm^3, that reacts to form 2.34 g of $NaCl$.

The reaction takes place at room temperature and pressure.

volume of chlorine gas = ... cm^3 [3]

(f) Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

(i) Explain why, in terms of structure and bonding.

....

....

....

.... [3]

(ii) Name the product formed at the positive electrode when electricity is passed through molten sodium chloride.

.... [1]

(iii) State the type of change that occurs at the positive electrode in **(ii)**.

Explain your answer in terms of electron transfer.

type of change ....

explanation ....
[2]

(iv) Describe what else can be done to sodium chloride to allow it to conduct electricity.

.... [1]

[Total: 26]

(d) When hydrated magnesium sulfate crystals, $MgSO_4{\bullet}xH_2O$, are heated they give off water.

$$MgSO_4{\bullet}xH_2O(s) \rightarrow MgSO_4(s) + xH_2O(g)$$

A student carries out an experiment to determine the value of x in $MgSO_4{\bullet}xH_2O$.

Step 1 Hydrated magnesium sulfate crystals were weighed.

Step 2 Hydrated magnesium sulfate crystals were heated.

Step 3 The remaining solid was weighed.

(i) Describe how the student can ensure that all the water is given off.

....

....

.... [2]

(ii) In an experiment, all the water was removed from 1.23 g of $MgSO_4{\bullet}xH_2O$. The mass of $MgSO_4$ remaining was 0.60 g.

M_r: $MgSO_4$ = 120; M_r: H_2O = 18

Determine the value of x using the following steps.

- Calculate the number of moles of $MgSO_4$ remaining.

moles of $MgSO_4$ = ...

- Calculate the mass of H_2O given off.

mass of H_2O = ... g

- Calculate the moles of H_2O given off.

moles of H_2O = ...

- Determine the value of x.

x = ...

[4]

[Total: 17]